FEB22009S Team PL4 Research Proposal: Battery Optimisation

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1. Research objective

We propose to develop two Integer Programming formulations to maximize household profit via optimal battery scheduling and grid arbitrage. Specifically, we adapt the Basic and Tighter Operation MIPs introduced by Elgersma et al. (2024) to rigorously model operational limits-including power boundaries, conversion efficiencies, and charge/discharge exclusivity-and evaluate their respective structural efficiencies.

2. Common model components

Let $\mathcal{T} = \{1, \ldots, T\}$ denote the set of time periods, where each period has length Δ hours.

Decision variables

For each $t \in \mathcal{T}$:

- $g_t^{\text{in}} \geq 0$: electricity bought from the grid (kW),
- $g_t^{\text{out}} \geq 0$: electricity sold to the grid (kW),
- $p_t^{\text{ch}} \geq 0$: battery charging power (kW),
- $p_t^{\text{dis}} \geq 0$: battery discharging power (kW),
- $s_t \geq 0$: battery state of charge at the end of period t (kWh),
- $\delta_t \in \{0, 1\}$: mode indicator, with $\delta_t = 1$ for charging and $\delta_t = 0$ for discharging.

Parameters

- d_t: net demand (kW), equal to consumption minus solar production,
- λ_t: electricity price (€/kWh),
- $S^{\min}, S^{\max}$: minimum and maximum state of charge (kWh),
- $P^{\text{C}}, P^{\text{D}}$: charging and discharging power limits (kW),
- $\eta^{\text{C}}, \eta^{\text{D}} \in (0, 1]$: charging and discharging efficiencies,
- S_0: initial state of charge (kWh).

Objective function

Household profit equals revenue from electricity exported to the grid minus the cost of electricity imported from the grid. Because g_t^{in} and g_t^{out} are measured in kW while λ_t is expressed in €/kWh, the energy traded in one period equals power times the period length Δ:

$$\max \sum_{t \in \mathcal{T}} \lambda_t \left(g_t^{\text{out}} - g_t^{\text{in}}\right) \Delta. \tag{1}$$

Power balance

Net demand must be balanced in every period:

$$g_t^{\text{in}} + p_t^{\text{dis}} - p_t^{\text{ch}} - g_t^{\text{out}} = d_t, \qquad \forall t \in \mathcal{T}. \tag{2}$$

Equation (2) allows both self-consumption and arbitrage: when prices are low, the model may buy extra electricity from the grid and store it in the battery for later use or resale.

3. Formulation 1: Basic Operation MIP

Following Section II.A of Elgersma et al. (2024), the battery dynamics are:

$$s_t = s_{t-1} + \eta^{\mathrm{C}} p_t^{\mathrm{ch}} \Delta - \frac{1}{\eta^{\mathrm{D}}} p_t^{\mathrm{dis}} \Delta, \qquad \forall t \in \mathcal{T}, \tag{3}$$

$$S^{\min} \le s_t \le S^{\max}, \qquad \forall t \in \mathcal{T}, \tag{4}$$

$$p_t^{\mathrm{ch}} \le P^{\mathrm{C}} \delta_t, \qquad \forall t \in \mathcal{T}, \tag{5}$$

$$p_t^{\mathrm{dis}} \le P^{\mathrm{D}} (1 - \delta_t), \qquad \forall t \in \mathcal{T}. \tag{6}$$

The initial condition is

$$s_0 = S_0. \tag{7}$$

Constraint (3) updates stored energy, where charging adds only $\eta^{\mathrm{C}} p_t^{\mathrm{ch}} \Delta$ kWh and discharging removes $p_t^{\mathrm{dis}} \Delta / \eta^{\mathrm{D}}$ kWh. Constraint (4) enforces the minimum and maximum battery capacity. Constraints (5)–(6) use the binary mode variable to prevent simultaneous charging and discharging.

4. Formulation 2: Tighter Operation MIP

The second formulation keeps the same objective (1), balance equation (2), initial condition (7), battery dynamics (3), and power limits (5)–(6). It replaces only the basic state-of-charge bounds (4) with the tighter inequalities from Section III.B of Elgersma et al. (2024):

$$s_{t-1} \ge S^{\min} + \frac{1}{\eta^{\mathrm{D}}} p_t^{\mathrm{dis}} \Delta, \qquad \forall t \in \mathcal{T}, \tag{8}$$

$$s_{t-1} \le S^{\max} - \eta^{\mathrm{C}} p_t^{\mathrm{ch}} \Delta, \qquad \forall t \in \mathcal{T}, \tag{9}$$

$$S^{\min} \le s_T \le S^{\max}. \tag{10}$$

Constraints (8) and (9) bound the *previous* state of charge instead of the end-of-period state of charge. More precisely, before discharging in period t, the battery must already contain enough energy to supply $p_t^{\mathrm{dis}} \Delta / \eta^{\mathrm{D}}$ on top of the minimum level, and before charging in period t, it must already have enough free capacity to absorb $\eta^{\mathrm{C}} p_t^{\mathrm{ch}} \Delta$ below the maximum level. These inequalities therefore cut off LP-relaxation solutions that are allowed by the basic formulation, while preserving the feasible integer solutions of the original storage operation problem.

5. Expected comparison and implementation plan

Both formulations model the same physical battery optimization problem and should therefore produce the same optimal *integer* solution. The difference is computational, the tighter formulation gives a stronger LP relaxation, so the solver is expected to explore fewer branch-and-bound nodes and solve faster, especially for longer time horizons. Moreover, as the electricity price may be negative, this may show differences between the Basic Operation and Tighter Operation models.

We will implement both models using the provided time-series data, and compare objective value, solve time, and relaxation strength. Further work will also consist of including in the model grid fees, battery degradation, and other additional extensions the team might find relevant. In the final report, this will form the basis for the later analysis of LP relaxations and computational performance.

References

Elgersma et al. (2024). *Tight MIP Formulations for Optimal Operation and Investment of Storage Including Reserves*. arXiv: `2411.17484`.